

Challenge -1

1) EC2 Ubuntu Instance Creation

- a. Region : ap-south-1
- b. Instance Name : Ec2-jenkins
- c. Key Pair : Abhishek.pem
- d. Security Key : default
- e. Screenshots for reference

Name and tags [Info](#)
Name
 [Add additional tags](#)

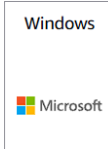
▼ Application and OS Images (Amazon Machine Image) [Info](#)
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

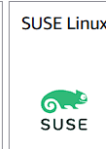

Amazon Linux


macOS

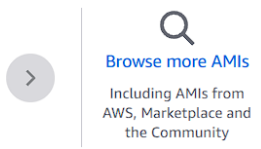

Ubuntu


Windows


Red Hat


SUSE Linux


Debian


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

▼ Instance type [Info](#) | [Get advice](#)
Instance type

t3.micro Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0108 USD per Hour On-Demand Windows base pricing: 0.02 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour

☒ All generations
[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
 [Create new key pair](#)

▼ Network settings Info

Edit

Network Info
vpc-00ce773080d81479f

Subnet Info
No preference (Default subnet in any availability zone)

Auto-assign public IP Info
Enable

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups Info
Select security groups ▼ [Compare security group rules](#)

default sg-0d93a1133c28252ee X
VPC: vpc-00ce773080d81479f

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Install Jenkins on EC2

Steps to Download Jenkins on EC2 Ubuntu Instance

1. `$ sudo apt update`
2. `$ sudo apt install fontconfig openjdk-21-jre -y`
3. `$ java -version`
4. Install the Jenkins repository key:
 - 4.1 `sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key`
 - 4.2 `sudo apt update`
5. Install Jenkins
 - a. `$ sudo apt install jenkins -y`
6. Start Jenkins
 - a. `$ sudo systemctl start jenkins`
 - b. `$ sudo systemctl enable jenkins`
7. Check Status of Jenkins
 - a. `$ sudo systemctl status jenkins`

```

ubuntu@ip-172-31-44-22:~$ sudo apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [515 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [120 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [175 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [55.5 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [430 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [84.8 kB]

```

```

183 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-44-22:~$ sudo apt install fontconfig openjdk-21-jre -y
Installing:
  fontconfig  openjdk-21-jre

Installing dependencies:
  adwaita-icon-theme  glycin-thumbnails  libavahi-client3  libgdk-pixbuf-2.0-0  libheif-plugin-aomdec  libpcsclite1  libxcb-dri3-0  libx16  luit
  alsa-topology-conf  gsettings-desktop-schemas  libavahi-common-data  libgdk-pixbuf2.0-common  libheif-plugin-aomenc  libpixman-1-0  libxcb-glx0  libx11-xcb1  mesa-libgl1
  alsa-ucm-conf  gtk-update-icon-cache  libavahi-common3  libgif7  libheif1  librsvg-2  libxcb-present0  libxcb-xfixes0  mesa-vulkan-drivers
  at-spi2-common  hicolor-icon-theme  libcairo-gobject2  libgl1  libhwy-tf64  librsvg2-common  libxcb-randr0  libxmu6  openjdk-21-jre-headless
  at-spi2-core ibus-gtk3  libcairo2  libgl1-mesa-dri  libibus-1.0-5  libsharpyuv0  libxcb-render0  libxpm4  user-session-manager
  bubblewrap  java-common  libcolor2  libglvnd0  libice6  libsm6  libxcb-shape0  libxrandr2  x11-common
  ca-certificates-java  libfontconfig1  libcolor2  libglvnd0  libjpeg-turbo8  libthai-data  libxcb-shm0  libxrender1  x11-utils
  dcconf-gsettings-backend  libasound2-data  libfontconfig1  libjpeg8  libthai0  libxcb-sync1  libxres1  libxshmfence1
  dcconf-service  libasound2t64  libfontconfig1  libjpeg8  libthai0  libxcb-sync1  libxres1  libxshmfence1
  fontconfig-config  libatk-bridge2.0-0t64  libdisplay-info3  libjpeg8  libthai0  libxcb-sync1  libxres1  libxshmfence1
  fonts-dejavu-core  libatk-wrapper-java  libdrm-intel1  libjpeg8  libthai0  libxcb-sync1  libxres1  libxshmfence1
  fonts-dejavu-extra  libatk-wrapper-java-jni  libepoxy0  libjpeg8  libthai0  libxcb-sync1  libxres1  libxshmfence1
  fonts-dejavu-mono  libatk1.0-0t64  libfontconfig1  libjpeg8  libthai0  libxcb-sync1  libxres1  libxshmfence1
  glycin-loaders  libatspi2.0-0t64  libgdm3  libjpeg8  libthai0  libxcb-sync1  libxres1  libxshmfence1

Suggested packages:
  adwaita-icon-theme-legacy  alsa-utils  gvfs  libheif-plugin-jpegdec  libheif-plugin-kvazaar  pcsd  fonts-ipafont-mincho  mesa-utils
  libheif-plugin-x265  libasound2-plugins  libheif-plugin-libde265  libheif-plugin-jpegenc  libheif-plugin-rav1e  librsync-bin  fonts-way-microhei
  | libheif-plugin-kvazaar  colord  libheif-plugin-x265  libheif-plugin-j2kdec  libheif-plugin-svtenc  libnss-mdns  | fonts-way-zenhei
  default-jre  cups-common  libheif-plugin-fmpgdec  libheif-plugin-j2kenc  liblms2-utils  fonts-ipafont-gothic  fonts-indic

Summary:
  Upgrading: 0, Installing: 121, Removing: 0, Not Installing: 183

```

```

ubuntu@ip-172-31-44-22:~$ java -version
openjdk version "21.0.12.1" 2026-08-18
OpenJDK Runtime Environment (build 21.0.12.1+1-1-26.04.4-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12.1+1-1-26.04.4-Ubuntu, mixed mode, sharing)

```

```

ubuntu@ip-172-31-44-22:~$ sudo apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
183 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-44-22:~$ sudo apt install jenkins -y
Package jenkins is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

```

```

ubuntu@ip-172-31-44-22:~$ sudo apt install jenkins -y
Installing:
jenkins

Installing dependencies:
net-tools

Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 183
Download size: 101 MB
Space needed: 102 MB / 4187 MB available

Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 net-tools amd64 2.10-2ubuntu1 [152 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.568.3 [101 MB]
Fetched 101 MB in 2s (51.5 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 89573 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-2ubuntu1_amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.568.3_all.deb ...
Unpacking jenkins (2.568.3) ...
Setting up net-tools (2.10-2ubuntu1) ...
Setting up jenkins (2.568.3) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' → '/usr/lib/systemd/system/jenkins.service'.
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

```

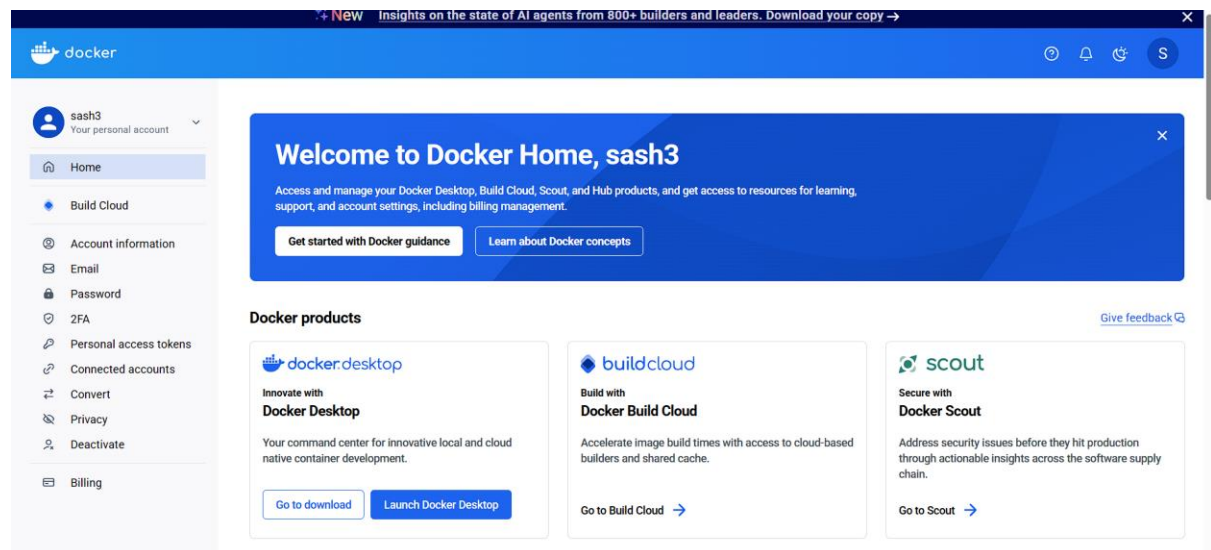
```

ubuntu@ip-172-31-44-22:~$ sudo systemctl start jenkins
ubuntu@ip-172-31-44-22:~$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
ubuntu@ip-172-31-44-22:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since wed 2026-09-23 16:37:07 UTC; 5min ago
     Invocation: 2cad884bda33447fbc9537259b0fe203
       Main PID: 4214 (java)
         Tasks: 37 (limit: 627)
        Memory: 308.2M (peak: 340.2M)
           CPU: 21.038s
      CGroup: /system.slice/jenkins.service
              └─4214 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Sep 23 16:37:01 ip-172-31-44-22 jenkins[4214]: [LF]>
Sep 23 16:37:01 ip-172-31-44-22 jenkins[4214]: [LF]>
Sep 23 16:37:01 ip-172-31-44-22 jenkins[4214]: [LF]>
Sep 23 16:37:01 ip-172-31-44-22 jenkins[4214]: [LF]>
Sep 23 16:37:07 ip-172-31-44-22 jenkins[4214]: 2026-09-23 16:37:07.194+0000 [id=32] INFO jenkins.InitReactorRunner$1onAttained: Completed initialization
Sep 23 16:37:07 ip-172-31-44-22 jenkins[4214]: 2026-09-23 16:37:07.218+0000 [id=24] INFO hudson.lifecycle.Lifecycle$onReady: Jenkins is fully up and running
Sep 23 16:37:07 ip-172-31-44-22 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Sep 23 16:37:07 ip-172-31-44-22 jenkins[4214]: 2026-09-23 16:37:07.892+0000 [id=50] INFO hudson.DownloadService$Downloadable$load: Obtained the updated data file for hudson.task
Sep 23 16:37:07 ip-172-31-44-22 jenkins[4214]: 2026-09-23 16:37:07.893+0000 [id=50] INFO hudson.util.Retrier$start: Performed the action check updates server successfully at

```

3.Setup Docker Desktop on your windows



Apache Tomcat on Ubuntu

- 1) `sudo apt update`
- 2) `sudo apt install openjdk-17-jdk -y`
- 3) `sudo useradd -m -U -d /opt/tomcat -s /bin/false tomcat`
- 4) `id tomcat`
- 5) `wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz`
- 6) `tar -xvzf apache-tomcat-11.0.24.tar.gz`
- 7) `sudo ls -l /opt/tomcat`
- 8) `sudo ls -l /opt/tomcat/apache-tomcat-11.0.26`
- 9) `sudo chown -R ubuntu:ubuntu /opt/tomcat`
- 10) `sudo chmod -R 755 /opt/tomcat`
- 11) `cd /opt/tomcat/apache-tomcat-11.0.26/bin`
- 12) `ls`
- 13) `sudo ./opt/tomcat/apache-tomcat-11.0.26/bin/startup.sh` (To start the tomcat service)

```
ubuntu@ip-172-31-34-55:~$ wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz
--2026-09-24 06:04:44-- https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14482233 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-11.0.26.tar.gz'

apache-tomcat-11.0.26.tar.gz 100%[=====] 13.81M ...KB/s in 0.05s
```

← → ↻ Not secure 13.53.212.158:8080 ☆ ⓘ All Bookmarks

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Apache Tomcat/11.0.26

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

Tomcat Setup	Realms & AAA	Examples	Servlet Specifications
First Web Application	JDBC DataSources		Tomcat Versions

Managing Tomcat

For security, access to the `manager.webapp` is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 11.0 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)

[Changelog](#)

[Migration Guide](#)

[Security Notices](#)

Documentation

[Tomcat 11.0 Documentation](#)

[Tomcat 11.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 11.0 Bug Database](#)
- [Tomcat 11.0 JavaDocs](#)
- [Tomcat 11.0 Git Repository at GitHub](#)

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [tomcat-dev](#)
Development mailing list, including commit messages